

Name: _____



Checking understanding of factors, multiples, squares and cubes

Task A: Multiples

1) Using the grid:

a) Draw a circle next to all the **multiples** of 2

b) Draw a square next to all the **multiples** of 3

c) Draw a triangle next to all the **multiples** of 5

d) What do the numbers with a circle and square represent?

e) What do the numbers with a circle, square and triangle represent?

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35

Task B: Factors

1) Only using the following cards once, fill in the gaps.

2

3

5

7

11

12

13

14

15

22

- a)

is a multiple of
- b)

is a multiple of
- c)

is a multiple of
- d)

is a factor of
- e)

only has factors

2) Are the following always true, sometimes true or never true? When the answer is sometimes or never true, give an example

	Always true	Sometimes true	Never true
Given A, B and C are integers $A \times B = C$ C is a multiple of A and B A and B are factors of C			
A multiple of a number is bigger than its largest factor			
A product of two integers is an integer			
1 is a factor of every integer			

Task C: Square and cube numbers

1) Work out the unknown number given the clues.

- a)

I am less than 100
 I am greater than 50
 I am a square number
 I am odd

b)

I am a cube number
 The sum of my digits is 9
 I am a multiple of 9
 I am less than 100

c)

I am a factor of 24
 I am a multiple of 2
 I am a number less than 10
 I am a square number

d)

I am a cube number
 I am a square number
 I am a factor of every number

2) Complete the Venn diagram showing all the square and cube numbers up to 200.

